

Stellar models & their stability

The differential equations of stellar evolution:

(i) $\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho}$: the mass distribution

(ii) $\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} - \frac{1}{4\pi r^2} \frac{\partial^2 r}{\partial t^2}$: Pressure gradient

in HE: $\frac{\partial^2 r}{\partial t^2} = 0$

(iii) $\frac{\partial l}{\partial m} = \epsilon_{\text{nuc}} - \epsilon_{\nu} - \epsilon_{\text{grav}}$: luminosity gradient

in TE: $\epsilon_{\text{grav}} = 0$

in expansion: $\epsilon_{\text{grav}} < 0$

in contraction: $\epsilon_{\text{grav}} > 0$

(iv) $\frac{\partial T}{\partial m} = -\frac{Gm}{4\pi r^4} \frac{T}{P} \nabla$: temperature gradient

$$\nabla = \begin{cases} \nabla_{\text{rad}} = \frac{3\kappa}{16\pi ac_2} \frac{lP}{mT^4} & \text{if } \nabla_{\text{rad}} \leq \nabla_{\text{ad}} \\ \nabla_{\text{ad}} + \Delta\nabla & \text{if } \nabla_{\text{ad}} > \nabla_{\text{rad}} \end{cases}$$

$\Delta\nabla$: 0 if inside star's outermost layers

$$\nabla_{\text{ad}} = \left(\frac{\partial \log T}{\partial \log P} \right)_{\text{ad}}$$

(v) $\frac{\partial x_i}{\partial t} = \frac{A_i m_H}{\rho} \left(-\sum_j (1 + \delta_{ij}) r_{ij} + \sum_{k,l} r_{kl,i} \right)$: evolution composition

(→) contains mixing terms in practice which can enhance the rate of interactions

Each of these equations are solved simultaneously in stellar evolution codes.

$\{P, \rho, K, \nabla_{\text{ad}}, \Delta v, \epsilon_{\text{nr}}, \epsilon_{\text{v}}, r_{ij}\}$ are all functions of ρ, T, X_i

→ $4+N$ unknown variables $\frac{1}{N}$ $4+N$ partial differential equations

Variables: r, ρ, T, l, X_i

→ functions of independent variables M, T

Timescales :

- $\frac{\partial^2 r}{\partial t^2}$ describes hydrodynamical changes that occur on τ_{dyn} timescales
→ almost always set to zero

- $-T \frac{\partial s}{\partial t} = \epsilon_{\text{gr}}$ describes thermal structure changes which occur on τ_{th} timescales

$-\tau \frac{\partial s}{\partial t}$: is sometimes set to zero.

- $\frac{\partial x_i}{\partial t}$ describes changes in composition which occur on the nuclear timescale.

→ these often decouple because
 $\tau_{\text{dyn}} \ll \tau_{\text{th}} \ll \tau_{\text{nuc}}$

★ In HE, TE these all become ODEs.

→ initial structure only depends on the initial composition

Boundary conditions:

$m=0$: $r=0$, $l=0$ at center.

at surface :

$m=M$, $r=R$

- simplest : $T=0$, $P=0$ @ surface

→ technically incorrect due to LSM

- better : surface is the photosphere which is defined as the region where photons escape → visible surface

The optical depth of the photosphere is

$$\tau_{ph} = \int_R^{\infty} dr \kappa \rho = \kappa_{ph} \int_R^{\infty} dr \rho = \frac{2}{3}$$

If the atmosphere is not a large fraction of the radius of the star (generally true),

$$\frac{dP}{dr} = -\frac{GM}{R^2} \rho \longrightarrow P(r) = \frac{GM}{R^2} \int_R^{\infty} dr \rho$$

also assume: $T(R) = T_{eff}$

$$\longrightarrow P(R) = \frac{GM}{R^2} \frac{2}{3\kappa_{ph}} ; L = 4\pi R^2 \sigma T_{eff}^4$$